10/5/2026

Denoising Diffusion Probabilistic Models
DDPM

Timeline:
- Before 2015: GANs were considered state of the art in image gen.

The unnoised / Lod process

Starting clear image: x_0

$x_0 \xrightarrow{q(x_1|x_0)} x_1$ — conditional distribution

adding gaussian noise:
$x_1 = x_0 + \beta \cdot \varepsilon$
$\varepsilon \sim N(0,1)$
$\rightarrow q(x_1|x_0) = N(x_0, \beta)$

$x_0 \rightarrow x_1 \xrightarrow{q(x_2|x_1)} x_2$

$x_2 = x_1 + \beta \cdot \varepsilon$

$q(x_2|x_1) = N(x_1, \beta)$

$\Rightarrow$

$x_2 = x_0 + 2\beta\varepsilon$
or
$q(x_2|x_0) = N(x_0, 2\beta)$

Generalizing to arbitrary step t:

$x_0 \rightarrow x_t$

$x_t = x_0 + t\beta \cdot \varepsilon$

so $q(x_t|x_0) = N(x_0, t\beta)$ — the variance explodes! "variance exploding diffusion"

→ However, the issue w/ this is that we want the diffusion process to converge to $N(0,1)$ b/c then all the original structures are gone + we know exactly what distribution to sample from when doing image gen.

Objective:

$q(x_t|x_0) \xrightarrow[t\to\infty]{} N(0,1)$

2 conditions:
① mean $\rightarrow 0$
② variance $\rightarrow 1$

this means that we must include a coeff. in front of the previous term to slowly decrease the mean

$\Rightarrow$

$x_{t-1} \rightarrow x_t$

$q(x_t|x_{t-1}) = \sqrt{1-\beta}\, x_{t-1} + \beta \cdot \varepsilon$

— Derivation skipped —

let $\bar{\alpha}_t = (1-\beta)^t$:

$q(x_t|x_{t-1}) = \sqrt{\bar{\alpha}_t}\, x_{t-1} + (1-\bar{\alpha}_t)\cdot\varepsilon$

If $\beta \in [0,1]$, $1-\beta \leq 0$ so $\bar{\alpha}_t \rightarrow 0$ as $t \rightarrow \infty$

Then, the mean $(\sqrt{\bar{\alpha}_t}\, x_{t-1}) \rightarrow 0$ and the variance $(1-\bar{\alpha}_t) \rightarrow 1$ as $t \rightarrow \infty$

NOTE: In practice, β changes between different stages, so $\bar{\alpha}_t = \prod_{i=1}^{t}(1-\beta_i)$

The variance preserving forward process:

$q(x_t|x_0) = \sqrt{\bar{\alpha}_t}\, x_0 + (1-\bar{\alpha}_t)\cdot\varepsilon$ where

$\bar{\alpha}_t = \prod_{i=1}^{t}(1-\beta_i)$

Useful Notation

derived from Bayesian statistics

joint probability = complete forward process: $q(x_1, x_2, \dots, x_T | x_0) = q(x_1|x_0)\, q(x_2|x_1, x_0) \cdots q(x_T|x_{T-1}, \dots, x_0)$

→ we can simplify b/c our forward process is a markov chain (x_t only depends on x_{t-1}):

$q(x_1, \dots, x_t | x_0) = q(x_1|x_0)\, q(x_2|x_1) \cdots q(x_t|x_{t-1})$

$q(x_{1:T}|x_0) = \prod_{t=1}^{T} q(x_t|x_{t-1})$

complete forward process:

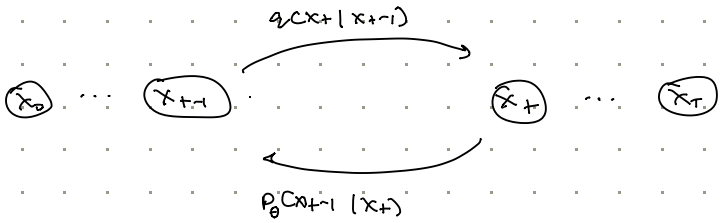
$q(x_{1:T}|x_0) = \prod_{t=1}^{T} q(x_t|x_{t-1})$

complete reverse process:

$p_\theta(x_{0:T}) = p_\theta(x_T) \prod_{t=1}^{T} p_\theta(x_{t-1}|x_t)$

Not conditioned b/c we start from pure Gaussian noise

The Reverse Process

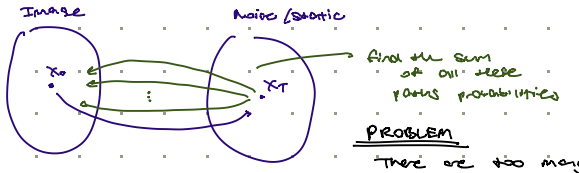


$p_\theta(x_{t-1} \mid x_t)$

↳ parameters we train in a neural network (weights) → Training the NN: minimize negative log-likelihood $-\log p_\theta(x_0)$

we have joint probabilities → marginalize distribution w.r.t other variables

$$-\log p_\theta(x_0) = -\log \int p_\theta(x_{0:T})\, dx_{1:T}$$



PROBLEM

There are too many paths → the quantity is intractable

To solve the intractable problem:

$$-\log p_\theta(x_0) = -\log \int q(x_{1:T} \mid x_0) \frac{p_\theta(x_{0:T})}{q(x_{1:T} \mid x_0)}\, dx_{1:T}$$

→ allows us to write an expectation

$$= -\log E_{q(x_{1:T} \mid x_0)} \left[\frac{p_\theta(x_{0:T})}{q(x_{1:T} \mid x_0)} \right]$$

↳ Jensen's Inequality

$$-\log p_\theta(x_0) \leq -E_{q(x_{1:T} \mid x_0)} \left[\log \frac{p_\theta(x_{0:T})}{q(x_{1:T} \mid x_0)} \right]$$

Evidence Lower Bound (ELBO)